

day-12-token-governor-100-closure

# Day 12 - Token Governor 100% Closure

Date: 2026-05-10

Branch: to-knowledge-mc

Lane: Token Governor

Status: DEVELOPER-SIDE CLOSED

Blocker class: NONE for the Day 12 preflight enforcement surface

Current verification HEAD: e0adce61d0e5f300abc69ad6569f3795fdb7486

Source implementation commit in current lineage: 0993942ad756a8ed1429c18c17459c7407dac199

Rollback for source implementation: git revert 0993942ad756a8ed1429c18c17459c7407dac199

## Scope

Day 12 closes the Token Governor preflight lane for the current Mission Control lineage. The accepted implementation already exists in the branch, so this closeout re-proves it on the current HEAD after the Day 11 dispatcher work.

This is not a claim that every model call in Mission Control is now hard-wired through Token Governor. That broader cross-route enforcement belongs to later dispatcher, status-consistency, and no-fake-action days. Day 12 acceptance is the working authenticated preflight route, tests, run time proof, audit persistence, and safe owner-facing blocker behavior.

## Current Implementation

- Canonical evaluator: 'src/lib/token-governor.ts'
- API route: 'src/app/api/gateway/token-governor/route.ts'
- Unit tests: 'src/lib/token-governor.test.ts'
- Route tests: 'src/app/api/gateway/token-governor/route.test.ts'
- Mounted designer page: '/gateway/token-governor' -> 'Token Governor.html'

The designer HTML/CSS/class names were not changed. The FULL v3 Token Governor mock remains the visual contract.

## Enforcement Behavior

The Token Governor preflight supports:

- 'allow' for requests within budget.
- 'warn' for requests at or above the warning threshold.
- 'block' for exhausted budgets.
- 'OWNER\_GATED' block for missing budgets.
- 'BLOCKED' block for restricted model/provider routes.

Known blocker tokens:

- 'token\_governor\_budget\_warning'
- 'token\_governor\_budget\_exhausted'
- 'token\_governor\_budget\_missing'
- 'token\_governor\_model\_restricted'

The preflight route never enables:

- execution
- writes
- external writes
- provider route changes

## Runtime Proof

Proof runtime:

- Base URL: 'http://127.0.0.1:3337'

- Runtime PID: 71076
- Listener: '127.0.0.1:3337'
- Public exposure: none observed in listener check
- Current HEAD during proof: 'e0adce61d0e5f300abc69ad6569f3795fdb7486'

Smoke results:

- 'GET /login': 200
- 'GET /api/gateway/token-governor' unauthenticated: 401
- 'GET /api/gateway/token-governor' authenticated: 200
  - 'canonical\_status': READY
  - 'blocker\_class': NONE
  - 'enforcement\_enabled': true
  - 'execution\_enabled': false
  - 'writes\_enabled': false
  - 'external\_writes\_enabled': false
  - 'provider\_route\_changes\_enabled': false
  - 'no\_secrets\_exposed': true
  - 'raw\_paths\_exposed': false

Preflight cases:

- Allow case:
  - HTTP 200
  - 'decision': allow
  - 'status': within\_budget
  - 'usage\_percent': 30
  - 'audit\_persisted': true
- Warning case:
  - HTTP 200
  - 'decision': warn
  - 'blocked\_reason': token\_governor\_budget\_warning
  - 'usage\_percent': 79
  - 'audit\_persisted': true
- Exhausted budget case:
  - HTTP 409
  - 'decision': block
  - 'blocked\_reason': token\_governor\_budget\_exhausted
  - 'usage\_percent': 91
  - 'audit\_persisted': true
- Restricted model case:
  - HTTP 409
  - 'decision': block
  - 'blocked\_reason': token\_governor\_model\_restricted
  - 'audit\_persisted': true

Audit proof:

- 'audit\_log' rows written for 'token\_governor.preflight' during this proof: 4

## UI Behavior

The Gateway Token Governor page remains mounted through the accepted FULL v3 designer mock:

- Route: '/gateway/token-governor'
- Mock file: 'public/designer-mission-control/design/gateway/Token Governor.html'
- Page source was not edited.
- Shared tokens, CSS classes, spacing, typography, and designer copy were not changed.

Current limitation:

- The designer mock is still static for visible Token Governor values.
- The live authenticated API is ready for later UI hydration, but Day 12 did not alter the mock because the owner explicitly directed that design files must remain the production visual contra

ct.

## ## Tests And Checks

Fresh checks run on 2026-05-10:

- 'pnpm test src/lib/token-governor.test.ts src/app/api/gateway/token-governor/route.test.ts'
  - 2 files passed
  - 9 tests passed
- 'git diff --check'
  - passed
- 'pnpm run typecheck'
  - passed
- 'pnpm run build'
  - passed
  - build route list includes '/api/gateway/token-governor' and '/gateway/token-governor'
- 'pnpm test'
  - 171 files passed
  - 1359 tests passed
- 'node scripts/check-protected-file-invariants.mjs'
  - ok: true
- 'node scripts/check-mission-control-route-rendering.mjs http://127.0.0.1:3337'
  - ok: true
  - routes checked: 46
  - designer pages checked: 8
  - failures: 0
- 'node scripts/check-button-contract-routes.mjs'
  - ok: true
  - missing route files: 0
- '.env' diff check
  - 0 bytes

Auxiliary harness note:

- 'scripts/check-button-contract-live-status.mjs' and 'scripts/check-protected-actions-locked.mjs' are not counted as Day 12 pass/fail evidence in this standalone runtime because they send only 'x-api-key'; this local runtime also requires the smoke session cookie for those protected authenticated probes and returns 401 without it.
- This is a later route-smoke harness compatibility issue, not a Token Governor enforcement failure. The Token Governor runtime proof above used the authenticated local smoke context.

## ## Safety Confirmation

- No '.env' changes.
- No secrets printed.
- No auth weakening.
- No fake send/upload/execution.
- No provider route changes.
- No external writes.
- No new public local exposure.
- No raw owner path exposure in API payloads.
- Parked duplicate artifacts remain untracked and unstaged.

## ## Remaining Blockers

None for the Day 12 Token Governor preflight acceptance gate.

Known future work:

- Later dispatcher/model-call days must wire every relevant provider execution path through Token Governor preflight before execution.
- Later UI days must hydrate visible Token Governor values without changing the accepted FULL v3 design contract.
- Later route-smoke days should update authenticated harness scripts so they can include the same

e local smoke session cookie used by the current runtime.

## ## Closeout Ledger

- Day number and lane: Day 12 - Token Governor
- Status: DEVELOPER-SIDE CLOSED
- Blocker classification: NONE
- Source implementation commit: '0993942ad756a8ed1429c18c17459c7407dac199'
- Current verification HEAD: 'e0adce61d0e5f300abc69ad6569f3795fdb7486'
- Push result for source implementation: already pushed in current branch lineage
- Refreshed report commit: pending
- Runtime proof: local-only standalone proof on '127.0.0.1:3337', pid 71076
- Rollback command: 'git revert 0993942ad756a8ed1429c18c17459c7407dac199'
- Next day automatically started: Day 13 - Mission Control Shell 100% Closure